Trigonometry

Section 1: Right Triangle Trigonometry (SOH-CAH-TOA)

1 In a right triangle ABC with right angle at C, the leg $AC = 8$ and the hypotenuse $AB = 17$. What is the value of $\tan B$?

a. $\frac{8}{15}$

b. $\frac{15}{8}$

c. $\frac{8}{17}$

d. $\frac{15}{17}$

2 If $\sin\theta = \frac{5}{13}$ and θ is an acute angle, what is the value of $\cos\theta$?

a. $\frac{12}{13}$

b. $\frac{5}{12}$

c. $\frac{8}{13}$

d. $\frac{13}{12}$

3 A 20-foot ladder leans against a vertical wall, making a 60° angle with the ground. How high up the wall does the ladder reach?

a. 10 ft

b. $10\sqrt{3}$ ft

c. 15 ft

d. $20\sqrt{3}$ ft

Section 2: Unit Circle & Angles in Standard Position

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4 What is the exact value of $\cos(\frac{5\pi}{6})$?

a. $\frac{\sqrt{3}}{2}$

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

5 If the terminal side of an angle θ in standard position passes through the point $(-3, -4)$, what is the value of $\sec\theta$?

a. $-\frac{5}{3}$

b. $-\frac{5}{4}$

c. $\frac{5}{3}$

d. $-\frac{3}{5}$

6 What is the exact value of $\tan(\frac{7\pi}{4})$?

a. 1

b. -1

c. $\sqrt{3}$

d. $-\frac{\sqrt{3}}{3}$

Section 3: Laws of Sines & Cosines

7 In triangle ABC, side $a = 10$, angle $A = 30^\circ$, and angle $B = 45^\circ$. What is the exact length of side b?

a. $10\sqrt{2}$

b. $5\sqrt{2}$

c. $10\sqrt{3}$

d. 15

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8 In triangle PQR, sides $p = 5$, $q = 8$, and the included angle $R = 60^\circ$. What is the length of side r?

a. 6

b. 7

c. 8

d. $\sqrt{49}$

9 What is the area of a triangle with side lengths 6 cm and 10 cm and an included angle of 150°?

a. $15\ \text{cm}^2$

b. $30\ \text{cm}^2$

c. $15\sqrt{3}\ \text{cm}^2$

d. $30\sqrt{3}\ \text{cm}^2$

Section 4: Trigonometric Graphs & Identities

10 What is the period of the function $f(x) = 4\sin(3x - \pi) + 2$?

a. $\frac{2\pi}{3}$

b. 3π

c. $\frac{\pi}{3}$

d. 2π

11 Which of the following expressions is identically equal to $\frac{\sin^2 x}{1 - \cos x}$ for all valid x?

a. $1 + \cos x$

b. $1 - \cos x$

c. $\sin x$

d. $\cos x$

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12 What is the amplitude of the cosine function $g(x) = -5\cos(2x + \frac{\pi}{4}) - 3$?

a. -5

b. 2

c. 5

d. 8

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